

Graphing data

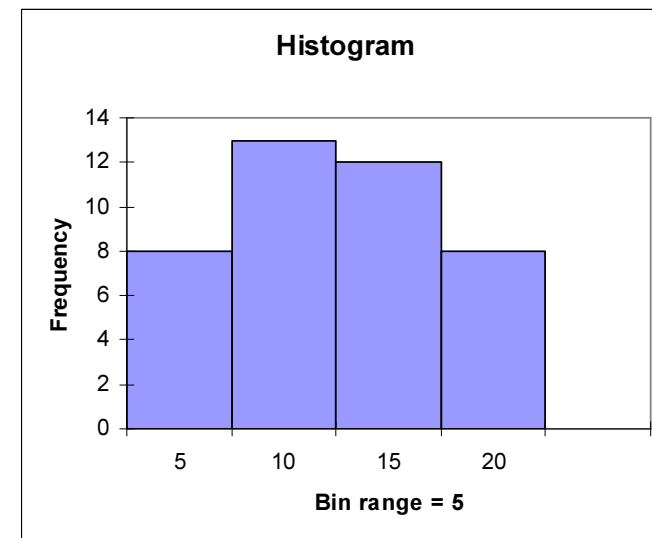
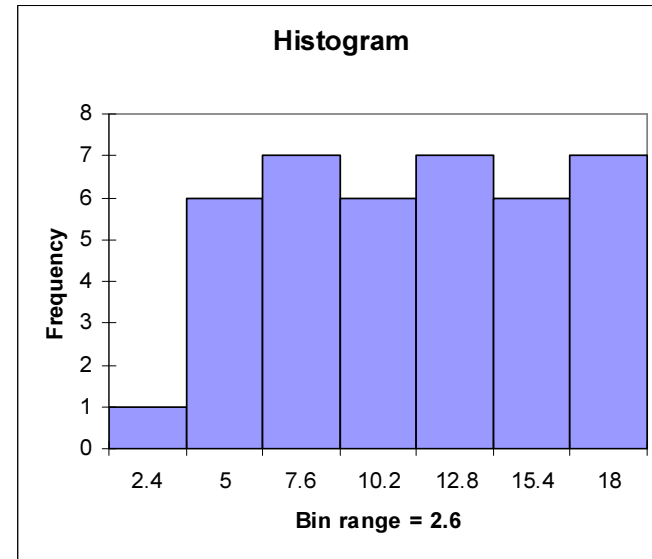
Graphical analysis techniques

When you collect data, the first thing you should do is plot it. Graphical methods help you understand the data by revealing characteristics such as the shape, mode, outliers, or data collection errors.

Once one of my students worked through a long homework problem, calculating the descriptive statistics and error values, but did not plot her data first. If she had, she would have seen one point that did not follow the sawtooth pattern of the data—a point whose value she had entered incorrectly. All her calculations were wrong.

Histograms

The histogram plots the frequency of grouped, or “binned” data. The shape of the histogram is significantly affected by the size and number of bins (see the two histograms at right, where the data are binned using two different range sizes). There’s no One Right Way to design histograms. One rule of thumb to determine the **number** of bins is to take the square root of the number of data points and round it either up or down. One method of determining the **size** of each bin is to divide the range of the data (Max-Min) by the number of bins.



Stem and leaf diagrams (1)

Stem and leaf diagrams show you the range of your data and the frequency of values. These diagrams can help determine the best grouping for histograms. The left-most significant digit is the “stem”, and the remainder is the “leaf”.

Suppose we have these data:

{85 69 69 74 51 85 81 96 64 84 77
78 117 73 73 85 87 84 95 84}

The left-most significant digit is in the 10's place.

Value	Stem	Leaf
85	8	5
69	6	9
69	6	9
74	7	4
51	5	1
85	8	5
81	8	1
96	9	6
64	6	4
84	8	4
77	7	7
78	7	8
117	11	7
73	7	3
73	7	3
85	8	5
87	8	7
84	8	4
95	9	5
84	8	4

Stem and leaf diagrams (2)

In the final diagram, we sort the stems in descending order and the leaves in ascending order.

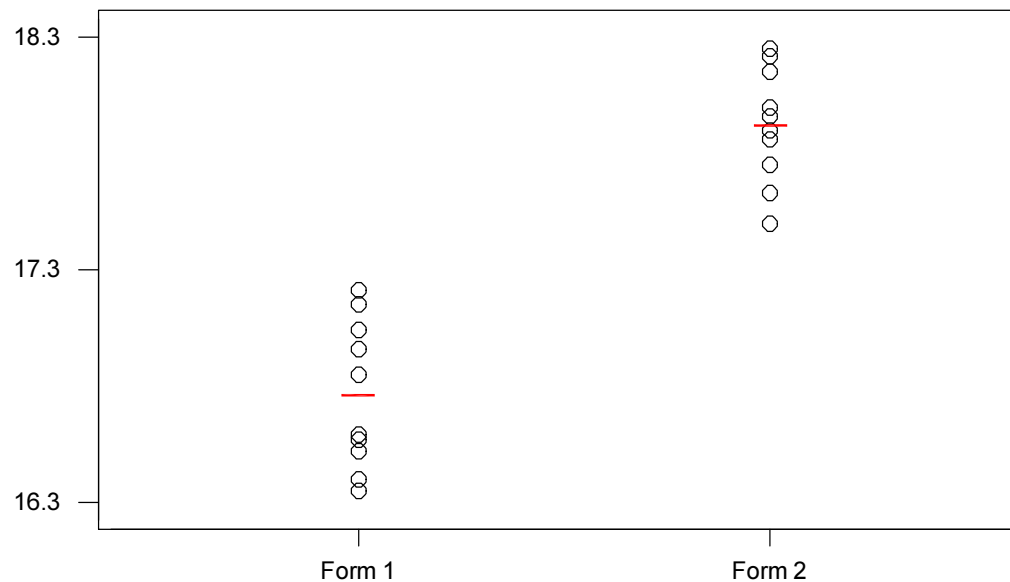
<u>Stem</u>	<u>Leaf</u>
11	7
10	
9	56
8	14445557
7	3378
6	499
5	1

Note that we included the “10” stem even though we don’t have a value in that range. Now we can clearly see the gap in the data.

If you rotate the stem and leaf diagram counterclockwise, it resembles a histogram.

Dot diagrams

These diagrams plot the data as dots in a single column. The mean is shown as a horizontal line. In this representation, the data are not grouped, or aggregated, as they are in a histogram. You can clearly see that there's a significant difference between the two sets of data.



Bar charts

These charts are similar to histograms; they compare frequencies or ranges of data groups.

The biggest mistake most people make when generating bar charts is to use a 3D chart for 2D data.

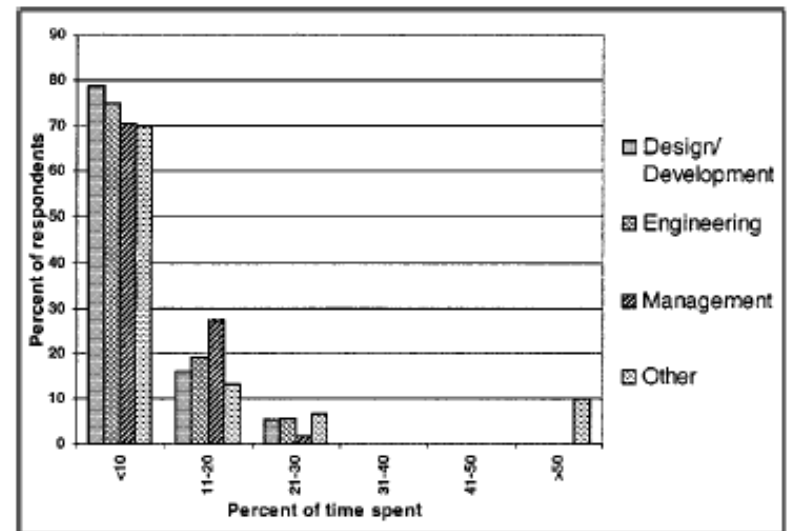
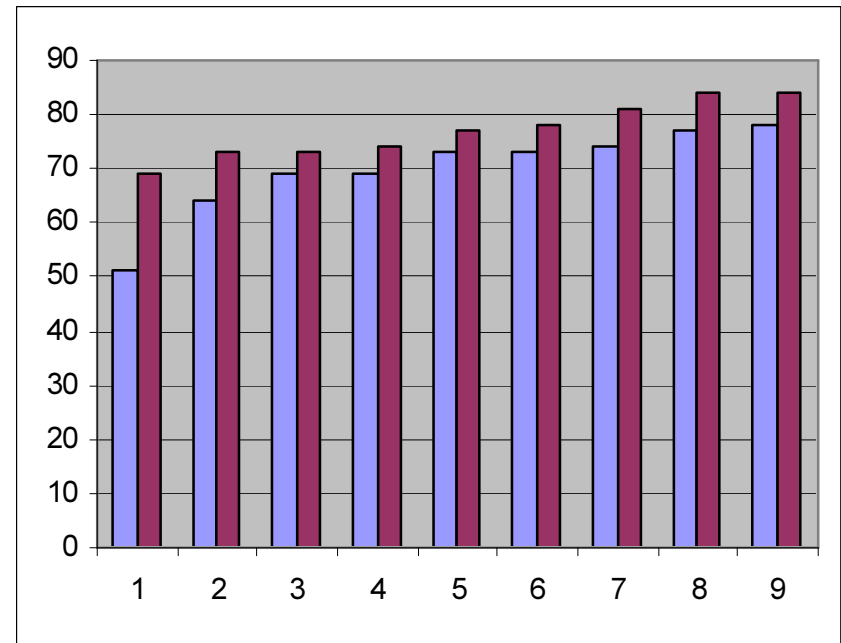
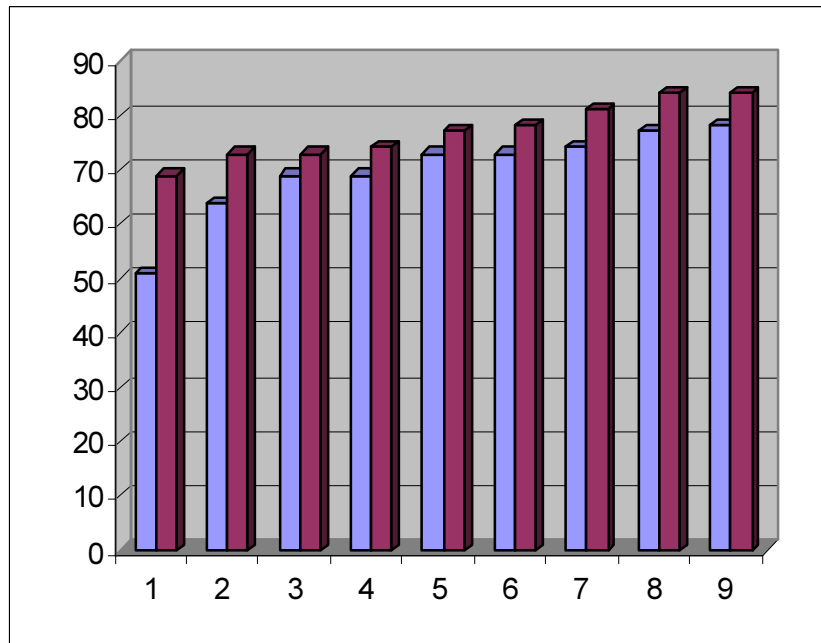


Figure 3. Relationship between the job category and the percent time spent on formal oral communication.

DON'T DO THAT!

3D vs 2D : using the same dataset

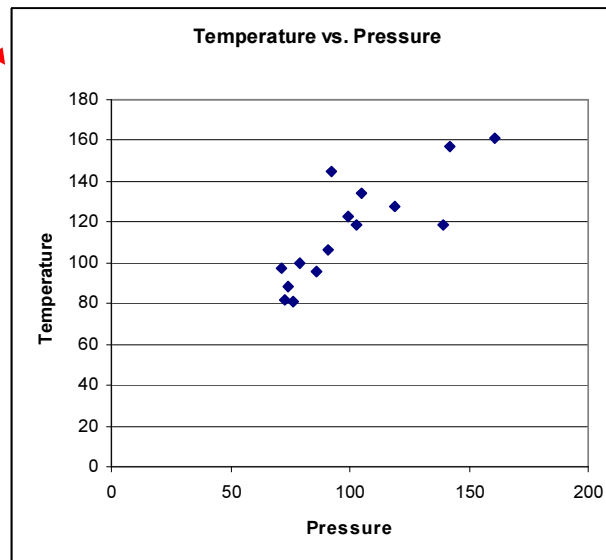
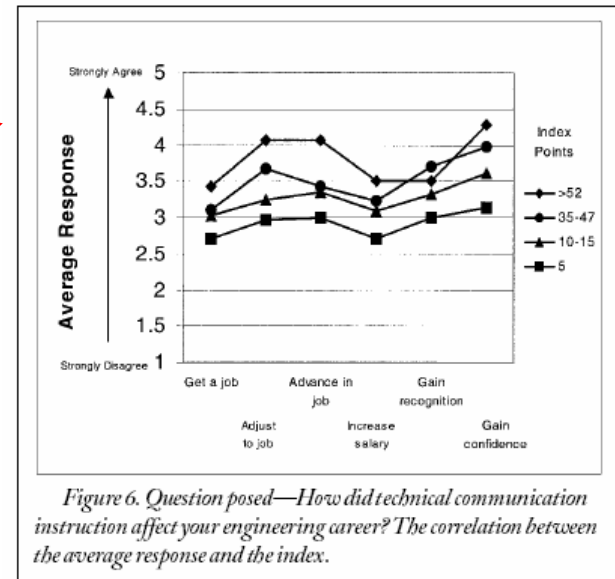


Which graph is easier to read?

* 2D means 2 dimensions to each data point. For this dataset, the first "dimension" is the order or position of the data point in the dataset; the second "dimension" is the value.

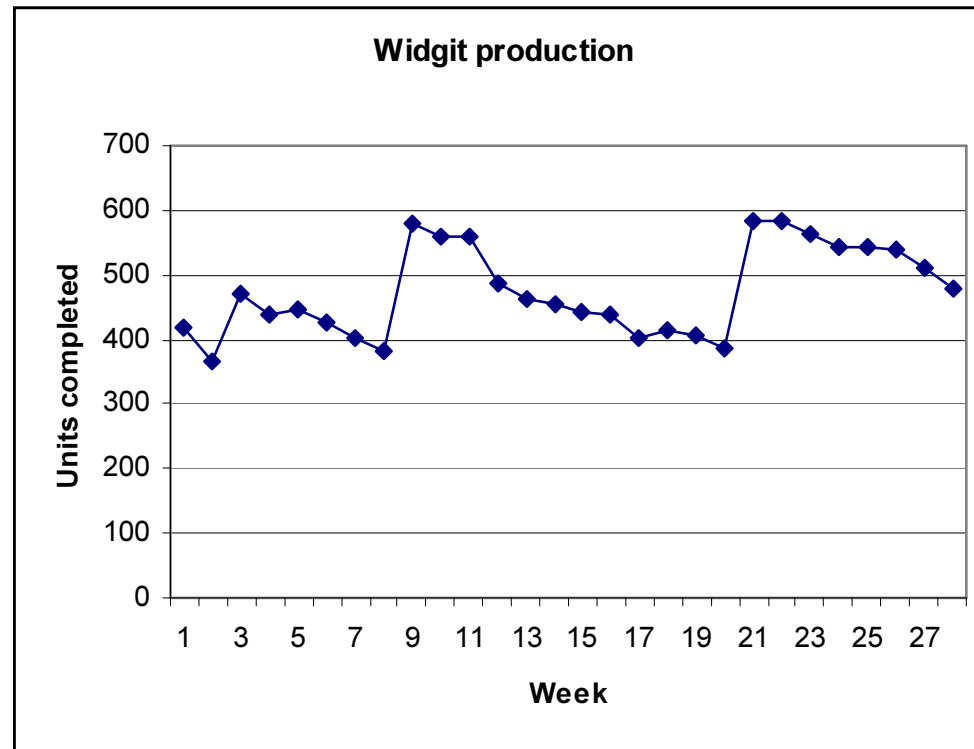
Line charts

These charts are may be very familiar to you. They may show **comparison among groups** or between two variables x and y , as in the case of **scatter charts**.



Line charts (2)

Or they may show values changing over time. These charts are often called **run charts**.



Pie charts

These charts are used mostly to show the ratio of variables to a whole, such as the distribution of personal income or the percent of time spent on subtasks.

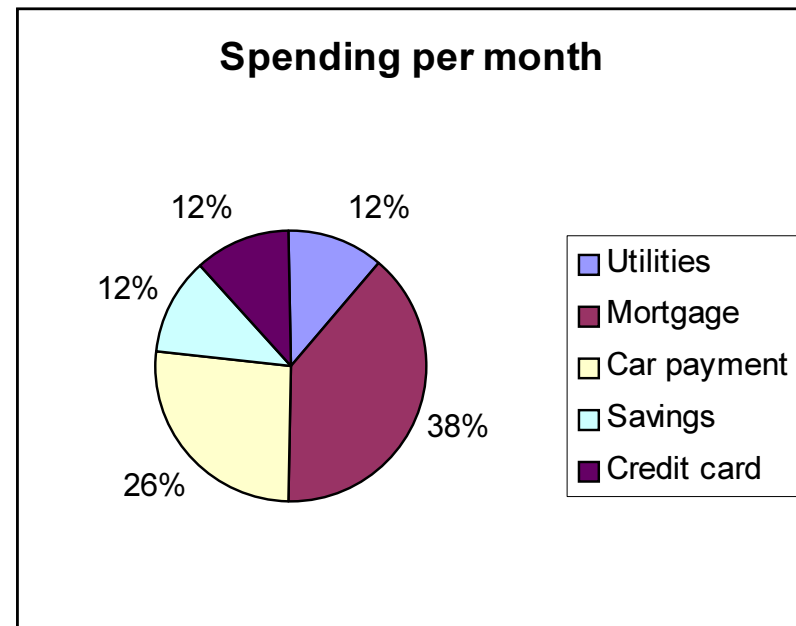
Make sure you are using this chart correctly. It should represent data in proportion to the entire dataset; the ratios **must** add up to 100% and the variables **must** be related to each other as part of a whole. A common mistake is to use variables that have no relation to each other—that are not parts of a whole.

Pie charts (2)

Pie charts are actually harder to “read” than bar charts.

The eye cannot discern the relative sizes of the slices as well as it can the relative heights of bars.

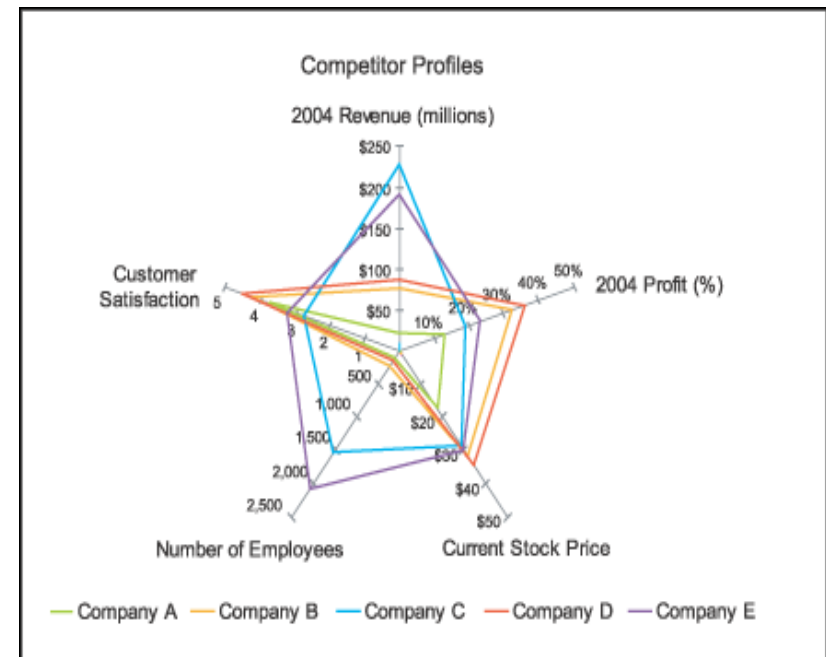
NEVER use 3D pie charts. The thickness of the 3D band distorts the size of the slices.



Radar charts (1)

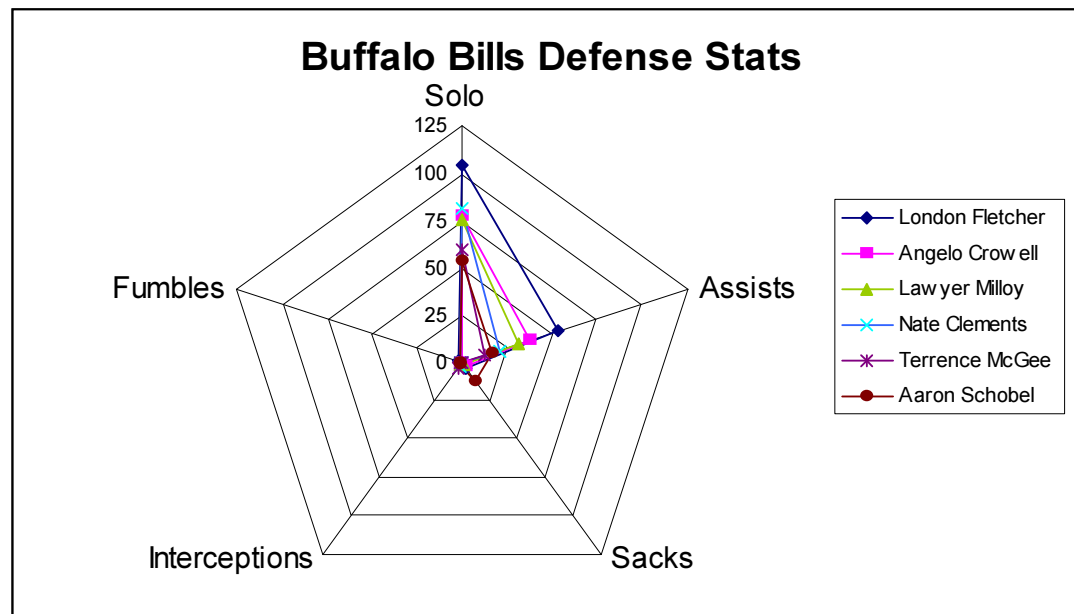
These charts compare several factors at once. The factors are laid out in a circle instead of linearly. Each factor can have its own axis scale, or one scale may be used for all axes. Points near the center are low; points near the edge are high.

This chart has different scales for each axis. But is this chart easy to read? Which company is “better”?



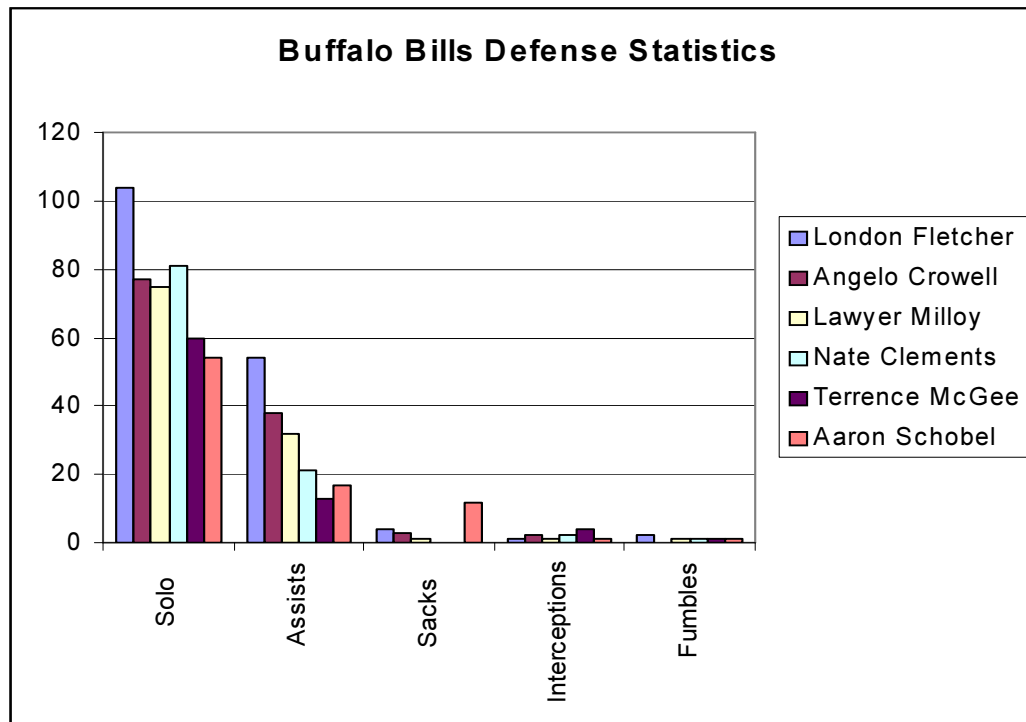
Radar charts (2)

In this chart, the same scale (frequency) is used for each factor. However, the scale is set by the number of solo tackles, and is too large for interceptions, fumble returns, and sacks—which are typically not as frequent as solo tackles and assists.



Radar charts (3)

A bar chart might be a better choice for this data. Solos and assists still dominate the scale, but in the remaining factors Aaron Schobel's sack numbers clearly stand out.



Radar charts (4)

“Readability” should always be an important consideration when choosing a graph type.

Radar charts might be a good choice when:

- Different scales are needed for each measure (as in the [Bills chart](#))
- You are trying to look at symmetry of data and not necessarily magnitude (as in the [company chart](#), you may want companies that are more like C and E and less like A, B, and D).
- The data are circular in nature or by convention (such as clock time data).

Pareto charts

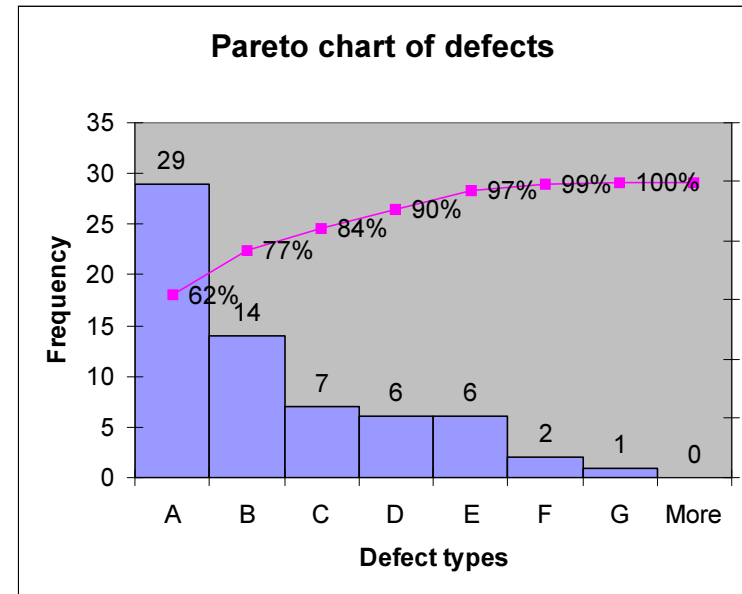
Another method of focusing on a problem solution is to generate a *Pareto chart*. These charts are based on a theory by Vilfredo Pareto, an economist who lived in Italy in the mid-1800s. Pareto found that 80% of the land in Italy was owned by 20% of the population. This 80/20 rule applies to many other areas, such as industry and manufacturing. In general, for example, 20% of stock generates 80% of inventory costs; 80% of production delays are caused by 20% of the possible causes.

Pareto charts, cont.

A Pareto chart ranks data categories in descending order of frequency, or percentage, or amount (however you are evaluating their effect).

The chart also includes a line that tracks the cumulative contribution of the ordered data categories.

We can use Pareto charts to determine which problems to concentrate on.



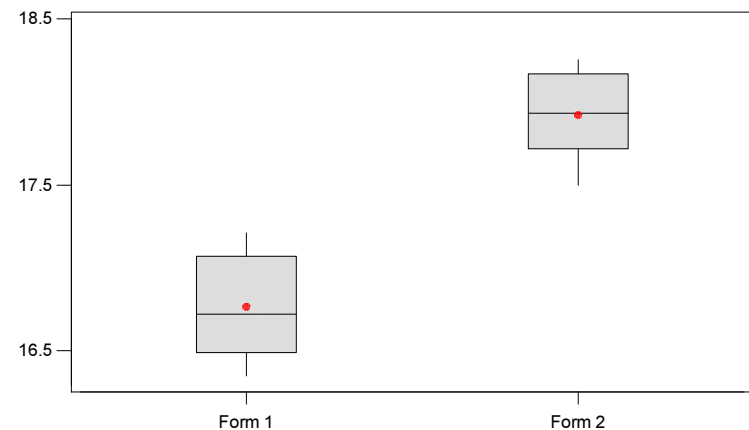
In this example, we would focus on defect types A, B, and C because they are responsible for ~80% of the defect frequency.

Box and whisker plots (1)

These plots were introduced by J. Tukey, a prominent statistician.

They are used to compare two or more sets of data (although they can be used for a single set).

To draw a box and whisker plot, you must find the mean, median, range, and upper and lower quartiles* of the data. We know how to get the mean and median—but how do we get the quartiles?



* Also called "percentiles"

Box and whisker plots (2)

An easy way is to take the median of the lower half of the data for the lower quartile Q1, then take the median of the upper half of the data for the upper quartile Q3.

Let's use this dataset:

{85 69 69 74 51 85 81 96 64 84 77 78 117 73 73 85 87 84 95 84}

Remember, to find the median we need to order the data.

{51 64 69 69 73 73 74 77 78 81 84 84 84 85 85 85 87 95 96 117}

The median for this data is 82.5.

Box and whisker plots (3)

{51 64 69 69 73 73 74 77 78 81 84 84 84 85 85 85 87 95 96 117}

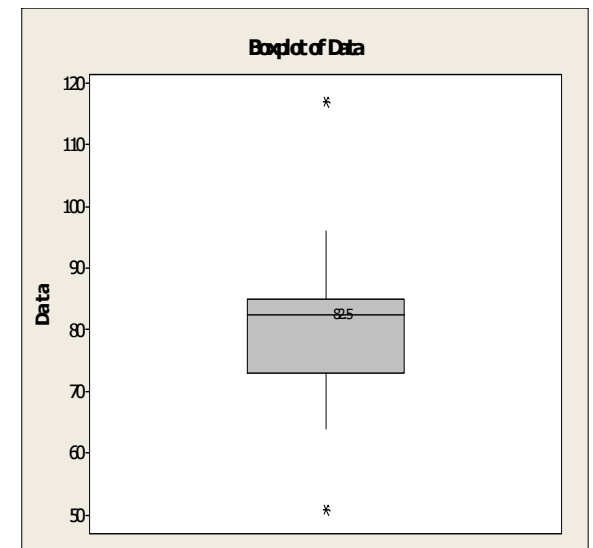
The green numbers are to the left of the median, and we use those to find the lower quartile. For these data, the lower quartile, Q_1 , is 73.

The blue numbers are to the right of the median, and we use those to find the upper quartile. For these data, the upper quartile, Q_3 , is 85.

Box and whisker plots (5)

So now we have all the information we need to make the box plots. The top of the box is the upper quartile value $Q3$, **85**. The bottom is the lower quartile value $Q1$, **73**. The line across the box is the median, 82.5.

The lines extending from the box are called *whiskers*. Some software plots the whiskers from the box to the upper and lower values. Minitab only extends the whiskers to an adjacent value.



Box and whisker plots (6)

The upper whisker is extended to the highest value within the range $\{Q3, Q3 + 1.5 (Q3 - Q1)\}$

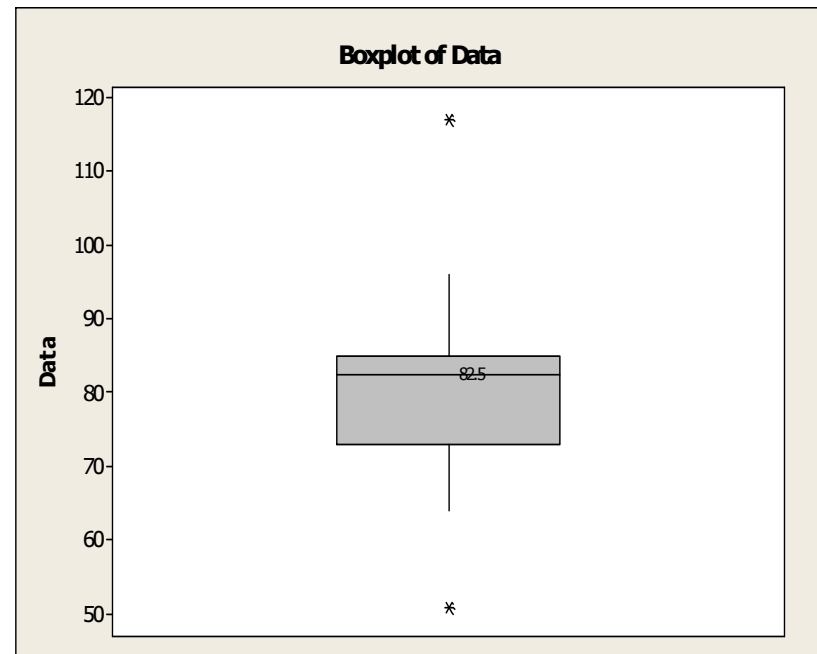
For our data, that range is from $\{85, 103\}$. The highest value from that range is 96.

Similarly, the lower whisker extends to the lowest value within the range $\{Q1 - 1.5 (Q3 - Q1), Q1\}$.

What's the range, and what's the value for our data?

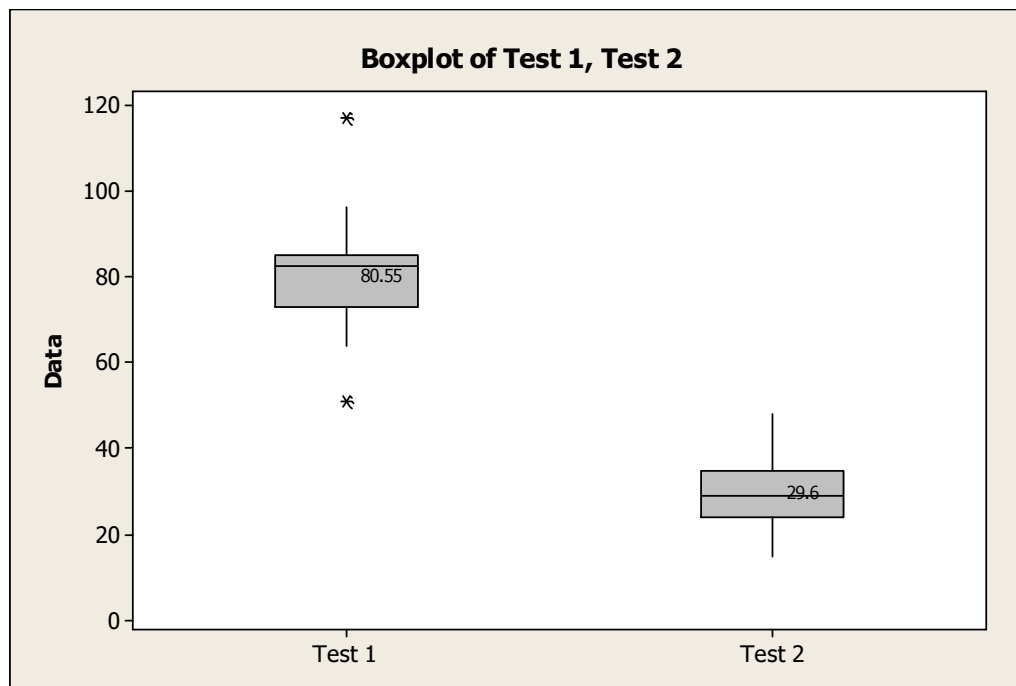
Box and whisker plots (7)

The highest value in our data, 117, and the lowest value, 51, are considered by Minitab to be outliers because they fall outside the range of the whiskers. These two values are marked with asterisks.



Box and whisker plots (8)

Some boxplots also label the mean. Here, Minitab uses the data value to show the mean. Sometimes you will see the mean shown as a dot inside the box.



Distribution shapes

Notice that the boxplots aren't always symmetrical. In the case of Test 1, the median is above the center of the box. That distribution is left-skewed. Most of the values appear in the higher end of the dataset's range. Conversely, a right-skewed distribution has more values in the lower end of the dataset's range.

